MTH1001-A

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

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Mathematical Structures

Module Leader: Dr Gihan Marasingha

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No word count

SECTION A

1. (a) (i) What is the name of the principle that states $x = x$ for every term x?

   (ii) State the substitution principle.

   (iii) Use the substitution principle to prove that if a and b are terms of the same type such that $a = b$, then $b = a$. (10)

   (b) In this question, select **one or more** possible deductions from a list of premises. You need not provide reasoning.

   (i) Given the premises P and $\neg Q$, which of the following can be deduced: P, Q, $P \wedge Q$, $P \vee Q$, $\neg P$?

   (ii) Given the premises $\neg P$ and Q, which of the following can be deduced: P, Q, $P \wedge Q$, $P \vee Q$, $\neg P$?

   (iii) Given the premise $P \vee Q$, which of the following can be deduced: P, Q, $P \wedge Q$, $P \vee Q$, $\neg P$?

   (iv) Given the premises $P \to Q$ and P, which of the following can be deduced: P, Q, $P \wedge Q$, $P \vee Q$, $\neg P$? (8)

   (c) (i) Let A and B be types. Let $f : A \to B$ be a function. Give the definition of 'f is surjective'.

   (ii) The functions f and g are defined as follows.

   $f : \mathbb{N} \to \mathbb{Z}$, where $f(x) = 3x$,
   $g : \mathbb{Z} \to \mathbb{N}$, where $g(y) = |y + 1|$.

   Which of f and g are surjective? In each case give a proof of your answer. Here, $\mathbb{N}$ is the type of natural numbers (non-negative integers) and $\mathbb{Z}$ is the type of integers. (10)

   (d) Let $A = \{2, 4\}$, $B = \{1, 5, 9\}$, $C = \{1, 4, 9\}$, $D = \{2, 5\}$ be sets on the type $\mathbb{N}$. Write each of the following sets as lists of elements enclosed in braces **and** state the number of elements in each set. You need not provide reasoning.

   (i) $A \cap C$, (ii) $B \cup C$, (iii) $B \setminus C$,
   (iv) $\mathcal{P}(D)$, (v) $B \times D$, (vi) $\mathcal{P}(D \setminus A)$. (12)

   (e) Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers. Let a be a real number.

   (i) Define what it means for $(a_n)_{n=0}^{\infty}$ to converge to a.

   (ii) State the Archimedean property of $\mathbb{R}$.

   (iii) Let $(a_n)_{n=0}^{\infty}$ be the sequence given by $a_n := 5 + 1/n^2$. Using the definition of limit and the Archimedean property of $\mathbb{R}$, prove that $(a_n)_{n=0}^{\infty}$ converges to 5. (10)

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SECTION B

2. (a) Let A and B be propositions. Prove that $(B \vee \neg A) \to (A \to B)$ using only the introduction and elimination rules of logic. (12)

   (b) Let S be a set of real numbers.

   (i) Give the definition of an *upper bound* of S.

   (ii) Give the definition of a *supremum* of S.

   (iii) Let A and B be sets of real numbers. Given that A and B have suprema, does it follow that $A \setminus B$ has a supremum? Justify your answer.

   (iv) Let U and V be non-empty sets of real numbers. Suppose that V has a supremum. Given that $\forall u \in U,\ \exists v \in V,\ u \leq 5v$, does it follow that U has a supremum? Justify your answer. (13)

   [25]

3. (a) Let A and B be types, let S and T be sets of elements of type A and let $f : A \to B$ be a function.

   (i) Give the definition of the image $f(T)$.

   (ii) Prove that $f(S \cup T) = f(S) \cup f(T)$. Use only the definitions of set operators, the rules of logic, and theorems of equality. (13)

   (b) A binary operator $\boxdot$ is defined on $\mathbb{N}$ recursively by the rules:

   - for every $m : \mathbb{N}$, $m \boxdot 0 \coloneqq 1$,
   - for all $m, n : \mathbb{N}$, $m \boxdot S(n) \coloneqq (m \boxdot n) \times m$.

   By induction on b, prove that $m \boxdot (a + b) = (m \boxdot a) \times (m \boxdot b)$, for all $m, a, b : \mathbb{N}$. You may use all familiar properties of natural number addition and multiplication. (12)

   [25]

4. (a) Let A and B be types. Let $f : A \to B$ and $g : B \to A$ be functions.

   (i) What does it mean for g to be an *inverse* of f?

   (ii) Let $h : B \to A$ be a function. Prove that if g and h are inverses of f, then $g = h$. (10)

   (b) Let $(a_n)_{n=0}^{\infty}$ be a sequence of non-negative real numbers. Let $(a_{f(n)})_{n=0}^{\infty}$ be a subsequence of $(a_n)_{n=0}^{\infty}$.

   (i) What does it mean for $\sum_{n=0}^{\infty} a_n$ to converge?

   (ii) Given that $\sum_{n=0}^{\infty} a_n$ converges, prove, using the definition of convergence of series, that $\sum_{n=0}^{\infty} a_{f(n)}$ converges. You may use any result from lectures on the convergence of sequences. (15)

[**25**]